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A New Lease of Life for Thomson's Bonds Model of Intelligence

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Modern factor analysis is the outgrowth of Spearman's original "2-factor" model of intelligence, according to which a mental test score is regarded as the sum of a general factor and a specific factor. As early as 1914, Godfrey Thomson realized that the data did not require this interpretation and he demonstrated this by proposing what became known as his "bonds" model of intelligence. Van der Maas et al. (2006) have recently drawn attention to what they perceive as difficulties with both models and have proposed a 3rd model. Neither alternative requires the general factor that was at the core of Spearman's idea. Although Thomson's model has been largely forgotten, the authors show that it merits further consideration because it can compete, statistically and biologically, on equal terms with Spearman's model. In particular, they show that it is impossible to distinguish statistically between the 2 models. There are also lessons to be learnt from the way in which Thomson arrived at his model and from the subsequent debate between Spearman and Thomson. The extent to which the recent proposal by van der Maas et al. may offer any advantage over Spearman's and Thomson's models is unclear and requires further investigation.

Keywords: bonds, factor analysis, intelligence, Charles Spearman, Godfrey Thomson

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1. Background

Spearman's theory of two factors has been the basis for modeling intelligence since its introduction in 1904 (Spearman, 1904). According to this theory, or model as we call it, test scores are accounted for by supposing that each score is the sum of a general factor, which Spearman called *g*, and an independent specific factor. He called the general factor *g* to avoid the connotations of the more obvious term general intelligence, as the discussions reported in Deary, Lawn, and Bartholomew (2008) make clear. Spearman's model not only provided a plausible account of why test scores tend to be positively correlated but it was easily generalized, principally by Thurstone (1935), to account for more than one factor. Since then it has provided the launching pad for the modern theory of structural relations modeling in which there are assumed to be linear relationships between the latent variables.

The model has not been without its critics, most recently van der Maas et al. (2006), whose proposal we consider in a later section. However, it is not widely known that one of Spearman's contemporaries, Godfrey Thomson (Thomson, 1916), pointed out that the two-factor model was not the only way of accounting for the pattern of correlations commonly found between mental test scores. He proposed what we here call his *bonds model* as an alternative. The debate between Spearman (1916) and Thomson that this proposal initiated continued for more than 30 years. The matter was never conclusively resolved, but, for a variety of reasons, Spearman's model came to dominate subsequent developments.

This is an appropriate time to review the situation, for four reasons. First, as just noted, a third model has been proposed by van der Maas et al. (2006), which is capable of explaining the positive manifold without reference to a general factor. Second, there have been great advances in knowledge of how the brain works since Spearman's time, and it is now possible to make a more realistic assessment of the biological merits of the competing models. Third, a large amount of historical material has become available to us that illuminates the debates which took place between Spearman and Thomson and their supporters (e.g., Bartholomew, Deary, & Lawn, 2009a, 2009b). Finally, there have been advances in statistical modeling that make it possible to show that the Spearman and Thomson models can compete on more equal terms than had, hitherto, been recognized. In particular, as we shall show, there is no statistical means of distinguishing between the two models. The purpose of the present article is, therefore, to turn the spotlight onto Thomson's bonds model and to argue that it now merits a new lease of life.

The two-factor model accounted for two features often exhibited by the correlations between sets of mental test scores. The first was

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that they were usually all positive, indicating that good performance in one test tended to be positively correlated with the score in any other. Today, this pattern is sometimes referred to as the positive manifold. The second was a little more subtle. Spearman noticed that such matrices often displayed a particular pattern, which was described as hierarchical. This meant that the tests could be arranged in an order such that those close together in the sequence were more highly correlated than those more distant. More specifically, he noticed that the matrices of correlation coefficients had a particular structure. This can be described in various ways, but the one that became most popular was to say that the tetrad differences (defined below) were close to zero. Spearman was then able to show that the occurrence of such matrices could be explained by supposing that each test score was the sum of two parts: a part that was common to all tests and a part that was specific to the particular test being considered.

Any serious competitor to Spearman's model must be able to account for these two empirical facts. It is also highly desirable that it should be capable of being extended to cope with more than one general factor and that it should be consistent with contemporary understanding of how the brain works. We shall show that Thomson's bonds model is capable of passing all these tests.

2. Terminology

Before proceeding we need to bridge the time gap of more than a century by making some comments on terminology and notation.

For the first half of the twentieth century there was no well-developed concept of a *probability model*. We wish to speak of Spearman's two-factor *model*, but he called it a *theory*. It is similar with Thomson's *bonds*, or *sampling* model as it was often described. Opinions differed on what to call the elements that were at the core of Thomson's model but, as we shall see, Thomson eventually settled on the term *bond*, which he borrowed from his friend E. L. Thorndike (see Deary et al., 2008).

The term "factor" was used rather freely at this time and this could give rise to misunderstanding. Spearman spoke of his two-factor theory, using the term "factor" in the modern sense of factor analysis. But Mackie (1928, 1929) used the term for what Thomson later called a bond. However, in 1925 (Brown & Thomson, 1925, p. 188 and p. 194) Thomson was still using the term "factor." A general factor, then as now, was a factor that was presumed to exert an effect on every test item. Likewise, a specific factor influenced items in one specific test only. Between the two lay group factors that were presumed to have influence within a subset of tests of items. Group factors were associated with skill domains, like arithmetical ability, which would be called into play by only some of the items. Initially, group factors were central to the bonds theory, but, as we shall see, they quickly receded into the background.

There was a need to name quantities that appeared in the equations. Frequently these were called "qualities," but a distinction was made by careful writers, such as Garnett, between the thing itself and the measure used. As in modern test theory, we speak of items or test scores.

Much of the early discussion about the structure of correlation matrices turned on what were known as the *tetrad differences*, though the term has fallen out of use as the need for them has disappeared. These are most easily defined by reference to Spearman's two-factor model, considered in the next section, in connection with which they played a prominent role.

3. Spearman's Two-Factor Model

Spearman proposed that test scores could be regarded as made up of two parts. This may be expressed this by the equation

$$\text{Test score} = \text{effect of common factor} + \text{effect of specific factor.} \quad (1)$$

Today we should write it as

$$x_i = \lambda_i f + e_i \; (i = 1, 2, \ldots . n), \quad (2)$$

where x_i is the test score on the ith item, f is the common factor, and e_i is the specific factor for the ith item. For convenience, the xs, f, and the es are assumed to have zero means and f has unit variance. The multiplier, λ_i then arises because of our (implied) decision to give f unit variance. In modern practice the specific factor is usually thought of as an error term rather than a fixed quantity, a point to which we return later.

For a model with the structure given in Equation 2 the correlation between x_i and x_j has the simple form, $\lambda_i\lambda_j$. The tetrad differences are used to detect this structure.

A tetrad is a set of any four correlation coefficients, taken from each of the four corners of any rectangular array in the correlation matrix. The tetrad difference is the difference in the product of pairs from opposite corners. It is easy to check that all tetrad differences are zero if the model of Equation 2 applies. In practice, as Thomson and his contemporaries well knew, the empirical tetrad differences would rarely, if ever, all be precisely zero and a great deal of effort was expended in determining just how close to zero they needed to be—but that is not relevant to our present purpose. It is important, however, to realize that the test of zero tetrad differences was only one way of checking whether two models were equivalent. Another way was to calculate the correlation coefficients between columns of the correlation matrix. Today it is much easier to compare correlation structures directly, as we do in Section 9.

We next examine the alternative models proposed by Thomson (1916) and by van der Maas (2006).

4. Thomson's Bonds Model

As early as 1914 Thomson challenged Spearman's view that the two-factor theory was a necessary consequence of finding matrices with hierarchical structure, but publication was delayed, at Thomson's request, because of Spearman's war service. Thomson (1916) did this by demonstrating empirically that matrices of the required form could be generated by an entirely different set of assumptions. As we shall see, this idea was refined and given mathematical clothing throughout Thomson's lifetime.

It is important to be clear about what Thomson was actually claiming. He did not say that Spearman's model was wrong, but rather that it was not the only model that accounted for the data. Much of Thomson's later practical work was, in fact, predicated on the Spearman model and he certainly believed that one could proceed *as if* the factors were real. In other words, he had no objection to proceeding as if the factors were real. He put his

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position succinctly (Thomson, 1916, and Thomson, 1919, p. 408) by saying that the Spearman model was a *sufficient* but not *necessary* condition for the existence of real latent factors—a point he was to repeat tirelessly in years to come.

Thomson's thinking obviously developed over the years, but his central argument remained unchanged. Toward the end of his life he looked back on this as one of his main contributions to factor analysis, and it is to the final edition of his book (Thomson, 1951) that we must look for the clearest statement of the essential idea. He does not go into any technical detail but clearly restates, by means of an example, how two models can give rise to identical correlation matrices. In his own words, on p. 309, "The alternative theory to explain the zero tetrad-differences is that each test calls upon a sample of the bonds that the mind can form, and that some of these bonds are common to two tests and cause their correlation." After years of debate it was necessary to be clear about what exactly was meant by a *bond*. On p. 313 he wrote,

> What the "bonds" of the mind are, we do not know. But they are fairly certainly associated with the neurones or nerve cells of our brains, of which there are probably round about ten thousand million in each normal brain. Thinking is accompanied by the excitation of these neurones in patterns. The simplest patterns are instinctive, more complex ones acquired. Intelligence is possibly associated with the number and complexity of the patterns that the brain can (or could) make.

This model became known as the *all-or-none* version of the model because it assumed that the whole contribution from any bond was used.

5. The Mutual Interaction Model

Nothing we have said so far rules out the possibility that there may be further models, yet to be discovered, which are equally good. Recent work by van der Maas et al. (2006) suggests a totally different way of accounting for the positive correlations that one finds when analyzing mental test results. The aim of these authors was more modest in that they did not try to account for zero tetrad differences, which they did not mention, but only for the hierarchical structure of what they called the positive manifold of intelligence, that is, the tendency for all correlations to be positive.

Their model is actually a large family of models among which some, at least, produce the positive manifold. They argued that the brain contains a very large number of what they called *cognitive processes*. These are reminiscent of Thomson's bonds in the more general sense but, to avoid possible confusion, we use the terminology of van der Maas et al. (2006). Each cognitive process is characterized by a "value" that grows over time. In addition, there is a mutual interaction, or reciprocal causation, between processes, which means that the growth of one can influence that of others. The authors were able to demonstrate that, if interacting growth is governed by processes like those commonly occurring in ecological studies, for example, then there will be positive correlations among the values.

The key step in yielding a credible model is to link these values with the test scores that a subject would obtain, and this is where we have some difficulty. This requirement involves an implicit assumption that, to any possible test, there corresponds a unique cognitive process. This is the implication of the statement (van der Maas et al., 2006, p. 844) that, "We assume that for each process in our model, such a score is available. Given the large number of intelligence tests and the large number of cognitive tasks, this is feasible in principle." The most obvious interpretation of this is that the brain is equipped with one process for every possible test item-type that might be encountered. This seems inherently implausible to us. An alternative interpretation is that a cognitive process is assembled from single entities when the need arises. If so its value would, we believe, have to be some kind of composite value derived from the constituent entities. In that case it is not clear whether the growth and mutual interaction, which the model assumes to take place, are supposed to operate on the constituent elements or on the aggregated process. In the latter case it is not clear how this could happen when cognitive processes are constantly forming and then re-forming. They would have no fixed identity and so could not be described by the model as formulated. However, if composite processes are formed by selection from a larger set, then the similarity to Thomson's model is obvious.

Although van der Maas et al. (2006) did not make it a requirement of their model that it should yield zero tetrad differences, which Thomson and Spearman thought important, they did mention (in their footnote 10) that their models could give rise to special cases that would have this property. They also enumerated a number of respects in which they thought that Spearman's and Thomson's models fell short of what modern empirical data require. On the other hand, one obvious feature of their own models, which they share with Thomson's, is that they relate only to the case of one latent variable. In their case this is no drawback because their focus was on the measurement of the single latent variable designated "intelligence." However, the subsequent success of Spearman's two-factor model was partly due to the fact that it could be easily generalized, as Thurstone (1935) showed, to the case of several latent variables. Although the original bonds model suffered from the same drawback, we show in Section 9 that it can easily be generalized in the same way. It is not clear whether the same can be done for van der Maas et al.'s model.

We are urging that the bonds model merits further consideration, not because it is perfect, but because it captures the key features found in the data at least as well as its competitors.

6. Where Did the Idea for the Bonds Model Come From?

Although Thomson was very clear, from the beginning, that Spearman's model was not the only way of explaining zero tetrad differences, it has long puzzled us as to where the idea for his own model came from, because the idea of sampling bonds seems far removed from Spearman's idea of a common factor. If we could answer the question at the head of this section we might gain some insight into what advantage Thomson saw in his new idea. Fortunately, the new historical material available to us provides a partial answer. At the very outset, in 1914, Thomson felt that Spearman's model need not be the only explanation and so he began experimenting with dice-throwing experiments to explore the matter more closely. It was these results that he reported in his early papers. However, the insight he was lacking came to him quite suddenly. As he reported in his autobiography, published posthumously (Thomson, 1969),

> I well remember the most vivid, most emotionally moving, and I think most important flash of "illumination" that ever came to me, many

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> years ago now, in late 1918. Like most such (see Graham Wallas, or read Poincare) it came when I was not engaged consciously with the problem it concerned. I was in fact in our kitchen, busy with some sort of handyman repair job.

He goes on to say that it was only years later that a student of his in Edinburgh, Dr. John Mackie (see later), was able to express the result in mathematical form. Because of the timing—he had already published his original alternative to Spearman's theory (Thomson, 1916)—this illumination must have been related to the way in which the model accounted for tetrad differences rather than the basic idea for the model itself. In trying to reconstruct Thomson's line of thought it is useful to bear in mind the state of factor analysis at the time.

When it became clear that the two-factor model was not always adequate, Spearman, and others, resorted to introducing what were called "group" factors. This was suggested by the practical context of intelligence testing since it was easy to identify abilities which, it could be supposed, were tapped by subsets of tests. In the early work, therefore, a great deal of attention was given to determining whether it was necessary to introduce group factors. It was natural for Thomson to be thinking in these terms as he developed and defended his bonds model. On the other side of the Atlantic, however, Thurstone (1935) had generalized Spearman's model in a different way, by simply adding further factors without requiring them to have the specific form required by group factors. This proved to be the path that the development of factor analysis could take, and it provided the perspective in which these things are viewed today. Thurstone did, however, advocate rotation to what he called "simple structure," where the rotated factors loaded on a subset of tests and so, in effect, were group factors.

At this distance in time it is impossible to be sure how Thomson's mind was working, but the following attempt at reconstructing his line of thought has some plausibility and seems consistent with the evidence. Initially, Thomson must have had the insight to see that there must be other ways of explaining Spearman's zero tetrad differences, but at that point he evidently had no more than an intuitive idea of how this might come about. He simulated group factors by adding together the outcomes of several throws of a die. (These sums would have been approximately normally distributed by the central limit theorem, but Thomson did not note this.) He then simulated the specific factors in the same way. Each simulated test score was then supposed to be the sum of a group factor and a specific factor. The value of the group factor would, as its name implies, be common to several test scores, but each test score would have its own specific factor.

In this way Thomson created artificial test data generated according to his idea that there was no general factor. This was not sufficient for his purposes, because he knew that there would have to be some overlap in the group factors because, if there was not, test scores in one group would be uncorrelated with those in any other group and this would not lead to all tetrad differences being zero. So he arranged for the group factors to overlap. Two group factors overlap if there is at least one test score that is influenced by both of them. Thomson (1916) gives, in his Figure 1, the particular overlaps which he built into the experiment. He then generated data according to the model, computed the correlation coefficients, and thus showed that the tetrad differences were at least as close to zero as one would expect in a typical psychological experiment. Thus he refuted Spearman's claim that a general factor *must* be present.

It would be a simple exercise for the reader who wishes to gain insight into what Thomson was doing to repeat Thomson's (1916) sampling experiment using random numbers generated by a computer instead of rolling dice. As an alternative, we have set out the steps involved in simulating our modified model in sufficient detail, in Section 9 (and see the Appendix), to enable a simulation program to be written.

Spearman (1916) was not slow to point out the limitations of this demonstration. In the first place the particular pattern of overlap chosen seemed quite arbitrary, and second, the data had been generated by dice tossing and, if the whole experiment were to be repeated, a different result would be obtained. In fact, Spearman was at pains to point out, by a simple combinatorial argument, how many different ways there were of doing what Thomson had done. How could we know that they would all show the same pattern?

Thomson (1917/1919) gave a partial answer by remarking that the composition of the group factors could be chosen at random, and he gives such an example and shows that the same result was obtained. It was also in this paper that Thomson spelt out, for the first time, what his model actually was. The conclusion to the paper was a formal statement of the model, which we quote:

> The mind, in carrying out any activity such as a mental test, has two levels at which it can operate. The elements of activity at the lower level are entirely specific, but those at the higher level are such that they may come into play in different activities. Any activity is a sample of these elements. The elements are assumed to be additive like dice, and each to act on the 'all or none' principle, not being in fact further divisible. (p. 344)

The model expressed in these words parallels exactly the sampling scheme he had adopted earlier. The upper level of which he spoke corresponds to the area from which the group factors were selected. The elements (which later became the bonds) correspond to the dice and the contribution that each element made to the scores on the dice. The lower level generated the specifics in a similar manner. By the time of Brown and Thomson (1925, p. 195) it had become apparent to Thomson that the lower level was redundant, and he later claimed it as an advantage of his model that it was not really necessary to introduce specific factors at all (Thomson, 1939, p. 281).

Although the bonds were later interpreted as entities in the brain, it appears that they owed their origin to the fortuitous choice of group factors to generate his original samples. The important thing to notice is the introduction of random sampling. By selecting the bonds at random, the desired overlap in group factors would be achieved without incurring Spearman's criticism that they had been selected arbitrarily. In fact, in this transition, the group factors lost the significance they originally had and became merely artifacts of the sampling model.

It is important to note that, at this stage, the model was apparently arrived at on purely statistical grounds. As time passed, Thomson increasingly began to interpret it in biological terms by identifying bonds with entities in the brain.

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7. Generalizations: Mackie and Maxwell

As Thomson noted in his autobiography (Thomson, 1969), it was Dr. John Mackie who translated his "illumination" into precise mathematical form. This was done in Mackie (1929): Dodd (1929) provided a similar treatment at about the same time. The key idea was to suppose that any bond sampled contributed a certain amount to the test score. (In Thomson's empirical version of the model these contributions were numbers of points uppermost on the die.) The score was then the sum of the contributions from the selected bonds. This idea was translated into a more modern notation in Bartholomew (2007) and, because it appears as a special case of the more general treatment given in Section 9, we defer further discussion until that point.

A further generalization was made by Mackie (1929). One obvious extension was to suppose that a variable amount of a bond might be used instead of "all-or-none" as Thomson had supposed. Mackie supposed that specified proportions of the contribution from each bond are taken instead of the whole. He supposed that the proportions actually used were selected at random from all possible sets and calculated what the correlations between test scores would be under this assumption. He commented on two special cases. In the first he supposed that there was no sampling of bonds at all—all were used, and so the randomness arose solely from the uncertain proportions. In this case all of the correlations would have the value $2/\pi$. This is somewhat unreasonable because it is rare to find all correlations even nearly equal, much less actually equal to $2/\pi = .637$.

His second case is a combination of the all-or-none and the foregoing models in that bonds are first selected at random and then random proportions of their contributions are taken. In this case Mackie (1929) shows that the correlation between item *i* and item *j*, in modern notation, is

$$r_{ij} = 2\,\lambda_i\lambda_j/\pi \;(i, j = 1, 2, \ldots p). \tag{3}$$

The only difference between this and the result for the original bonds model is in the factor $2/\pi$. It is still true that the correlation is of the same form as required for agreement with Spearman's model. However, in this generalization it is impossible for the correlation to rise above a value of .637. In practice higher values may well occur, and Mackie had to rather arbitrarily suppose that correlations in excess of this value would require that one of the variables should be discarded, for being essentially the same as its partner. Such a generalization subtracts rather than adds to what the all-or-none model offered. Thomson, wisely no doubt, ignored this generalization in his later discussions of the model.

A second, and much more important generalization was made by Maxwell (1972). He was writing at a time when efficient methods were becoming available for fitting the factor model, and he was particularly concerned about what he saw as a misguided tendency to regard the factors obtained as real. In this article he attempted a double attack on this tendency. The first was a reminder that factors were not completely determined, but this does not concern us at present. The second was to point out that Thomson's bonds model provided a distinct alternative interpretation of the data and that this meant that one could not unambiguously interpret the data as though one of them were "true." However, by this time it was clear that more than one factor was often needed to fit data which were then available in intelligence testing. Thomson's model could not therefore be used as it stood, and Maxwell undertook to provide the generalization required to make his point.

Maxwell's (1972) idea was to assume that the supposed bonds were differentiated according to the type of item being attempted. Thus, for example, a subset of bonds might be specific to arithmetical items and another to spatial items. When such an item was attempted only this subset of items would be sampled. It turned out that such a scheme would give rise to data that could be fitted by a factor model, but one with several factors. Maxwell then took several data sets and showed how the analysis of each could be interpreted in terms of his extension of the bonds model. Maxwell's extension is, in fact, a special case of the more general model we give in Section 9, so we defer further details until that point.

8. The Debate Between Spearman and Thomson

Thomson never gave a satisfactory mathematical form to his model. As we have seen, his early papers on the subject were based on extensive sampling experiments (e.g., Thomson, 1916, 1919). He simply aimed to demonstrate that he could generate correlation matrices with something close to zero tetrad differences without invoking a general factor. As we have seen, it was Mackie who gave the model mathematical form, and this largely coincided with the treatment of Dodd (1929). According to the footnote on p. 43 of Thomson (1939), he broadly agreed with Dodd's treatment.

Thomson did, in fact, provide his own mathematical argument in support of his model (see, e.g., Thomson, 1951, p. 311). This is based on calculating the most likely numbers of bonds to be selected and then showing that these lead to zero tetrad differences. His argument seems to have been that the actual numbers of bonds will be close to the most likely and hence that the tetrad differences will not be too far from zero. Maxwell (1972) also used this approach without, apparently, noting that Mackie (1929) had given a fuller treatment.

Initially Spearman voiced the main objections to the bonds model, though their nature and content changed somewhat with time as Thomson developed his idea. Spearman (1916) started off by objecting to the arbitrariness of the selection of bonds. He also appeared to object to the element of chance in Thomson's model, because he did not think that chance could play any part in mental processes. It was too early for him to recognize that his own model would eventually be regarded as having a chance element through the presence of the "error" term in Equation 2. Spearman also argued that the bonds model allowed for no individual differences, because everyone was supposed to have exactly the same number of bonds. Thomson replied that it would not make any difference if the total number of bonds varied from one person to another.

More recent criticisms have been made, as detailed, for example, in van der Maas et al. (2006), which point to the failure of the theory to account for other empirically established facts, such as the very specific effects of brain damage on performance. We return to such matters in the final section, concentrating here on the main debate between Spearman and Thomson.

The most serious criticism, as Mackie (1928) clearly saw, was that Thomson's model was not really new but was simply Spearman's model in a different form. Spearman made this assertion in, for example, Spearman (1927, p. 96), but the justification for it lay

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in work by Maxwell Garnett. Spearman claimed, quoting Garnett's work in his support, that the specification of one model could be transformed into the other and hence that it was actually equivalent to the two-factor theory to which Spearman was firmly attached. Maxwell Garnett turned out to be a major figure in the dispute between Thomson and Spearman. He had been 16th wrangler at Cambridge in 1904 and worked in Karl Pearson's department at University College London, which, no doubt, was how he became acquainted with Spearman's work. Garnett was a formidable mathematician, and in a series of papers he demonstrated a masterly grasp of the distribution theory of normal variables. He used this knowledge in Spearman's service to demonstrate how a set of correlated normal variables, whose correlations had zero tetrad differences, could be expressed in terms of one general factor and a set of specific factors. He then went on to show how to transform a model of the Thomson kind into one closely resembling the two-factor theory. The "closely resembling" qualification is necessary because he had to allow the number of bonds to increase indefinitely—the larger the number of bonds, the nearer the tetrad differences would be to zero. (Had he used the slightly modified model we use, this would not have been necessary.) For all practical purposes this met Spearman's needs because he could now argue that the two models were, indeed, equivalent, and thus that Thomson's work was merely a restatement of his own.

Things were not quite that simple, however. Garnett was a very careful worker and, although his theoretical work was intricate, he was always very careful to say what he was doing and what he had actually proved. In this case (e.g., Garnett, 1919/1920) he made it very clear that, having shown the two models were equivalent, one had to make a recommendation as to which was preferable. On *mathematical* grounds it was clear that Spearman's model was much simpler and therefore to be preferred. Spearman read this as an endorsement of his own view. But this was not the real point, which was surely which model had the greater credibility in physiological terms, a point which he never addressed. For his part Thomson claimed that the advantage lay with his model because, as he put it, "it makes fewer and less special assumptions, because it is more elastic and wider and because it is in closer accord with theories in biology and in the study of heredity" (Brown & Thomson, 1925, p. 192).

We return to the biological point in Section 10. In reality, Garnett's result supported Thomson rather than Spearman because the former's model did relate to potentially identifiable entities in the brain, even if his account of what they were and how they worked was vague.

While this was going on, Mackie was examining the issue from Thomson's side. Mackie (1928) agreed that Garnett might have found a transformation taking one model into the other, but, he argued, the specific factors to which it led were not credible, because of the way in which they depended on the individual bonds.

This particular story culminated in a joint note by Garnett and Thomson (1919) which concludes, "we differ in the order of preference we assign to these alternative hypotheses, each of which, we agree, to be mathematically possible" (p. 368). This might have been the end of the story, but Maxwell Garnett returned to the fray giving what he believed was a further clarification of the issue between himself and Thomson and claiming that Thomson's results were consistent with Spearman's model (Garnett, 1919/1920). Thomson (1919/1920) could not let the matter rest and he responded almost immediately with a paper in which his exasperation is hardly disguised. He begins,

> It is with regret that I find it necessary, in view of a recent article by Mr. J. C. Maxwell Garnett, to return to the question of the phenomenon of hierarchical order among correlation coefficients and to insist, with all the emphasis of which my pen is capable, that the deduction of the presence of a General Factor which Professor Spearman has based upon this phenomenon is utterly and entirely invalid. (p. 319)

In this paper Thomson (1919/1920) reiterated the difference between sufficiency and necessity using, as an illustration, a polygon whose internal angles add up to 360° (four right angles). It is obviously necessary for a square to have this property, but having this property is not sufficient to make it a square. Thomson evidently thought that Garnett had not made this distinction and, it must be admitted, there is no evidence to contradict his view. However, there was a fundamental difference between the two which must have made it difficult for them to have come to terms with one another's thinking. Thomson thought in terms of examples using numbers and diagrams, whereas Garnett used a more rigorous algebraic approach. Thomson's (1919/1920) attitude to formalism is perfectly expressed by the following quotation from the same paper:

> I shall refrain from burdening the pages of this Journal with the mathematical symbols which would be required to prove this generally. Such a proof would be repellent to the non-mathematical reader, and is unnecessary to the mathematical reader, who can supply it for himself by analogy with the following simple illustration. (p. 322)

In relation to Garnett's own style he also says,

> Finally, if Mr. Garnett cares to try a numerical example (and his errors and misconceptions are almost entirely due to his neglect of this salutary practice) he will find that 13 group factors alone, with no specific factors at all, will give quite a reasonable hierarchy. (p. 325)

There is no such candid expression of Garnett's thoughts about this in his papers but, as Spearman's mathematical champion, he did express himself more frankly in correspondence with Spearman. Years later, when Garnett was with The League of Nations Union, he wrote, on February 8, 1927 (Garnett, 1927):

> My dear Spearman,
>
> It is most good of you to send me a copy of the manuscript of your reply to Thompson [*sic*]. Apart from your much too kind references to myself, I like your paper well. The principal trouble with Thompson is that he seems incapable of understanding any mathematical argument that is not quite simple. Since he finds it difficult to handle correlated variables in the abstract, he resorts to the concrete and makes use of dice. But he never enquires how this substitution of concrete dice for abstract variables affects his argument and his conclusions. As I explained to Mr. Dodd of Princeton on Sunday, Thompson's use of dice has the effect of substituting a bundle of lines for many-dimensional space with the result that any position not on one or other of the lines is, for Thompson, non existent. So he says that a general factor (represented by a direction in this many-dimensional space) does not exist if it is not identical with one of his lines; if one of the dice does not itself supply the general factor then there is, for Thompson, no general factor. All this amounts to no more

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than a materialisation for non-mathematicians of part of your admirable paper.

Yours sincerely, . . .

Fortunately, it is not necessary today to delve further into, or resolve, the differences between Thomson and Garnett because the whole matter can be expressed, and generalized, in contemporary terms. A bonds model in modern dress was given in Bartholomew (2007) and it is further extended in the following section.

9. A Modern and More General Version of the Bonds Model

As we have noted, Thomson never gave a precise mathematical statement of his model: This had to await the work of Mackie and Dodd. Here we propose a version of the model which, we think, captures the essence of what Thomson was after. However we also take matters further by showing that the idea can easily be extended to cover the case where there is more than one factor. This is interesting because, as we speculated in Bartholomew et al. (2009b), Thomson's model might easily be extended to give it the same generality as Thurstone (1935) achieved when he introduced multiple factor analysis as a generalization of the Spearman two-factor model.

Following Thomson we suppose there are N bonds in the brain which may be called into play when a person attempts a test item. If bond i is active, it contributes an amount e_i to the total score, which we denote by x_i. Mackie used the symbols $x, y, z, u \ldots$ to denote the quantities we are calling $e_1, e_2, e_3, \ldots$ but it is more economical to use a single symbol and distinguish the bonds by a subscript. It is important to state that this quantity e_i is a characteristic of the individual. If that bond is selected it will contribute that amount whatever the test item on all occasions. The values of the es are assumed to be determined by drawing them from a standard normal distribution. We assume the mean of this distribution to be zero and its standard deviation to be σ_i. This assumption involves no loss of generality because we also assume that the test scores have been standardized to have zero means.

The essence of Thomson's model was to suppose that a fixed proportion of bonds, p_i, say, are selected when test item i is selected. The resulting score would then be the sum of the es resulting from the sampled bonds. Thus if there were 10 bonds altogether and if $p_1 = 0.3$ then three bonds would be sampled at random; if these turned out to be Bonds 2, 5, and 7 then the first test score would be

$$x_1 = e_2 + e_5 + e_7$$

In general we may write

$$x_i = a_{i1}e_1 + a_{i2}e_2 + : : : + a_{iN}e_N, \tag{4}$$

where a_{ih} is an indicator random variable taking the value 1 if the hth bond is selected when the ith item is attempted and 0 otherwise. This equation may be compared with the corresponding equation for Spearman's in Equation 2. Thomson's sampling scheme implied that the es were mutually independent random variables with zero expectations and standard deviations σ_i; ($i = 1 \ldots, N$). This distribution describes the variation among individuals in the population from which the individuals are sampled.

At this point it is convenient to make a small alteration to the model as Thomson proposed it. This does not essentially change things but it avoids some tedious but unimportant complications which arise if we take a sample of fixed size. Instead of sampling a *fixed* number of bonds we suppose that each bond is selected independently with probability p_i, $i = 1 \ldots, n$, where n is the number of test items. The actual number of bonds will then be a random variable with the same expectation as in Thomson's version. In the case of the illustrative example given above, where we had $N = 10$ and $p_i = 0.3$, there would be a probability of 0.3 that any bond would be selected, independently of all others. The actual number selected would then be a binomial random variable but its expectation would be 3. Instead of having exactly three bonds selected we would have 3 on average.

The model expressed by Equation 4 can be expressed in matrix notation as

$$\mathbf{x} = \mathbf{A}\,\mathbf{e}, \tag{5}$$

where $\mathbf{x}$ is a column vector containing the xs, $\mathbf{e}$ is a vector of the es, and $\mathbf{A}$ is a matrix containing the indicator variables $\{a_{ij}\}$. This was the model considered in Bartholomew (2007), where it was shown that it led to a covariance matrix identical in form with that of Spearman's model. Here we propose a straightforward generalization which also has the same covariance matrix as the general linear factor model. It also includes Maxwell's (1972) extension as a special case.

The generalization that we propose is to suppose the final score is arrived at by repeating the sampling process, independently, several times. We refer to each such sampling process as a *pass* through the bonds because it is easier to think of them as happening sequentially, though this is by no means necessary. The first pass will be exactly as described above but the probability of selecting the bond i will now be denoted by p_{i1}, the 1 signifying that this refers to the first pass. At the second pass the probability of selection for each bond will change to p_{i2}. At the third pass it will be p_{i3} and so on up to the final, kth, pass when it is p_{ik}. The model says nothing about how many passes there are, just as in standard linear factor analysis the number of factors is not predetermined. The total score for any item will then will then be the sum of the contributions from the k passes.

If we let $\mathbf{x}$ denote the $n \times 1$ column vector of test scores the model may be written, in matrix notation, as

$$\mathbf{x} = \mathbf{A}_1\mathbf{e}_1 + \mathbf{A}_2\mathbf{e}_2 + \ldots + \mathbf{A}_k\mathbf{e}_k. \tag{6}$$

As before, the matrices $\mathbf{A}_r$ ($r = 1 \ldots, k$) contain indicator variables showing which bonds have been selected at each pass. Thus $a_{rih} = 1$ if item h is selected at pass r for item i, otherwise it is zero.

In the Appendix we show that this model leads to exactly the same covariances as the linear factor model with k factors. This means that there is no statistical way in which we can distinguish between the two types of model on the basis of their covariances alone. The practical usefulness thus turns on whether models belonging to the family given by Equation 6 have a plausible interpretation in terms of what happens in the brain. We already know that the special case when $k = 1$ corresponds to the bonds model as Thomson proposed it. It also includes the extension which Maxwell (1972) proposed, as we now show.

Maxwell (1972) suggested that some test items might call upon only a subset of the N bonds in the brain. At first sight this does not

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seem possible within the extension we are proposing, because this provides for a sampling from the whole set of bonds. But we can easily ensure that any bonds outside the chosen subset contribute nothing to the score. Suppose, for example, that Bond 10 is sampled, but that this is not in the chosen subset. Its contribution to the total score would be e_{10}, where this value is determined by drawing from a normal distribution with zero mean and standard deviation σ_{10}. But the value of σ_{10} can be anything we choose as long as the sum of the variances of the items selected remains the same for all tests. We can therefore make it zero and thus ensure that this item contributes nothing. In this way a test can be made to depend upon only a particular subset of the bonds. Any model of this kind will thus lead to something that is indistinguishable from a factor model as Maxwell showed. Consequently, the result of fitting any such model can be interpreted in terms of either model, as Maxwell demonstrated for his examples.

There are, of course, other variations on this theme, many of which may have no obvious interpretation, and this is an area which awaits further investigation. Nevertheless, the main point of the exercise has been to show that there is a broad family of models, including both the familiar factor model and the various versions of the bonds model, which cannot be distinguished by any statistical means. Any choice among them must therefore rest on biological considerations.

10. Spearman's and Thomson's Models of Intelligence and Brain Function

The motivation for Spearman's and Thomson's statistical models of intelligence was to account for the pattern of associations among diverse mental tests. Both speculated on how their statistical models might shed light on brain function. The concepts they employed were limited by the state of brain science at the times they were writing. In this reevaluation of their models—especially Thomson's, which has had far less attention—it is important to discuss their biological speculations, and then afterwards to enquire whether these are endorsed or contradicted by current findings on the biological basis of intelligence differences. Spearman (1927, pp. 117–135) wrote of *g* as being "mental energy," though he later partially revised this and stated that

> On weighing the evidence, many of us used to say that this G appears to measure some form of mental energy. But in the first place, such a suggestion is apt to invite needless controversy. This can be avoided by saying more cautiously that G behaves as if it measured an energy. In the second place, however, there seems to be good reason for changing the concept of energy to that of "power" (which, of course, is energy or work divided by time). In this way, one can talk about mind power in much the same manner as about horse power. (Monroe, 1931, pp. 156–157)

Thomson's (1939) assessment of the suggestion that intelligence might be equated with mental energy was that he worried that it was reification of a statistical finding and that, in any case, "there seems to be no easily conceivable way in which the 'energy' of the whole brain can be used in any direction indifferently, except by the 'neural engines' also taking part. The energy of a neuron seems to reside inside it . . ." (p. 241). In this case, there is partial support for Spearman. Thomson, in his 1939 book, did not discuss the importance of the mechanisms by which blood supply can be varied to different brain regions depending on mental demands. This is odd, because, 15 years earlier (Thomson, 1924, p. 232), he mentioned this explicitly and at some length as a possible biological contribution to Spearman's energy. Functional magnetic resonance imaging of the brain shows that, during mental work, blood flow increases to predictable specific regions of the brain involved in performing the task. Indeed, the principal indicator of neuronal activity during such brain imaging is the blood oxygenation level dependent (BOLD) signal, which is used to delineate those brain regions that are using more oxygen (extracted from hemoglobin in the adjacent blood vessels) than others. Therefore, to an extent, the energy of the brain may be used in any direction, as required. Using evidence from studies using the BOLD signal during mental work—and other perhaps better but less-used techniques such as positron emission tomography (PET)—the efficiency of a parieto-frontal brain system has been associated with intelligence (Jung & Haier, 2007).

A more obvious characteristic feature of the brain to which Thomson (and, perhaps, Spearman) might have turned in support of his ideas is its size. Larger brains, one might suppose, would contain more bonds and so lead to greater intelligence. We now know that people with higher intelligence test scores do, in fact, tend to have larger brains (McDaniel, 2005). Using magnetic resonance imaging to assess brain size in vivo in healthy subjects, one finds a significant but modest association ($r \sim 0{:}3$) between intelligence and overall brain size. To date, the evidence is that there is a distributed range of sites—including frontal, temporal, parietal, and occipital cortex—in which grey matter volume, especially, is associated with intelligence (Colom, Jung, & Haier, 2006; Haier, Jung, Yeo, Head, & Alkire, 2004). It is only recently, of course that this kind of measurement has been possible.

Thomson's "sampling of bonds[1]" theory (Thomson, 1939, p. 269) was based on his thought concerning brain biology that, "Instead of showing that the mind has a definite structure, being composed of a few factors which work through innumerable specific machines, the low rank shows that the mind has hardly any structure" (p. 270). He explicitly agreed with Thorndike's "quality hypothesis of intelligence that one mind is more intelligent than another simply because it possesses more interconnections out of which it can make patterns" (p. 281). His account of his theory repeats that there are no "faculties" and "organs" in the mind, which is "so comparatively structureless" (p. 280) compared with the body. He was not even certain that there were any identifiable elements: "It is possibly a continuum, its elements if any being more like the molecules of a dissolved crystalline substance than like grains of sand" (p. 271). But he did decide to offer the reader some conceptual elements: "For concreteness it is convenient to identify the elements, on the mental side, with something of the nature of Thorndike's 'bonds,' and on the bodily side with neurone arcs" (p. 271). (For more on the meaning of these terms refer back to Section 2.) Even early in his writing, Thomson (1919/1920) made some biological guesses as to the foundations of his elements

[1] Thomson uses "bond" repeatedly as a synonym for "element" or "unit" in his model of the brain's working. He might have used it first in Thomson (1924, p. 233), in which he stated that, "The great majority of bonds, I believe, are free on the whole from the defect of being, in some other situation, detrimental."

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that were to be sampled by mental tests, "If we attribute any degree of reality to the elements which enter into any mental test if, for example, we assume each to correspond to a neurone, or to a synapse, or to a nervous arc" (p. 326).

Thomson put few restrictions on the bonds: They might be inherited or caused by education; their actions were not necessarily additive; he allowed different readers to associate them with different entities; and he stated that "All a 'bond' means, is some very simple aspect of the causal background" (Thomson, 1939, p. 271). He did suggest that there might be "no linkages among the bonds" (p. 274) and that bonds came into the pattern required to solve test items in an all-or-none fashion. However, later on Thomson (1939) stated that

> When properly measured by a wide and varied hierarchical battery, g appears to him [Thomson] to be an index of the span of the whole mind, other common factors to measure only sub-pools, linkages among bonds. The former measures the whole number of bonds; the latter indicate the degree of structure among them. (p. 283)

The addition of subpools to his theory caused Spearman (1939a, 1939b) and Burt (1939a) to comment critically in their correspondence around the time of Thomson's (1939) book's first appearance. Burt (1940, p. 11, note 1) thought that Thomson's pulling back from his view that the mind was structureless, and to admitting to subpools, brought him closer to Spearman. Spearman (1939b) stated that the subpools were, "extremely obscure, both statistically and psychologically." One surprising comment from Burt to Spearman was that, "Of course, I fully agree that, as you say, the two theories are *not really rivals*. Both forms of analysis may quite well be possible simultaneously" (emphasis in the original). Both agreed that the Spearman–Thomson theories must be judged by "congruence with further observations." Thus far, it might seem as if Thomson's model of the brain has little that would contradict Spearman's view, but Thomson (1939) adds that

> The difference in point of view between the sampling theory and the two-factor theory [of Spearman] is that the latter looks upon g as being part of the test, while the former looks upon the test as being part of g. The two-factor theory is therefore compelled to postulate specific factors to account for the remainder of the variance of the test, and has to go on to offer some suggestions as to what specific factors are—perhaps, neural engines. The sampling theory simply says that the test requires only such and such a fraction of the bonds of the whole mind—the same fraction which, on the two-factor theory, g forms of the variance of the test. For it, specific factors are mere arguments, which do not arise unless, as can be done, the equations which represent the tests are so manipulated that there appears to be only one link connecting them all. The sampling theory does not make this transformation of the equations . . . . Those who do so, if they adhere to the interpretation that g means all the bonds of the whole mind, have to suppose that the whole mind first takes part in each activity, but that in addition a specific factor is concerned; which specific factor, since they have already invoked the whole mind, must be for them a second action of part of the mind annulling its former assistance—which is absurd. The two-factor equations then do not allow us to consider g as being all the bonds of the mind. (pp. 281–282)

Thomson (1939), despite this support for his own model, thought that Spearman's two-factor theory was likely to be more popular:

> The sampling theory lacked the good selling point of the other, but is comparatively free from its dangers, and seems much more likely to come into line, in due time, with physiological knowledge of the action of the nervous system. (p. 282)

The point also occurred in Brown and Thomson (1925, p. 190). Thomson, of course, was arguing within the context of intelligence testing, but his expectation was fulfilled, partly because factor analysis found applications in many other fields where the factors represented latent variables that had nothing to do with the brain. In the intelligence context, Thomson pleaded his case, leaving biology to be the arbiter. Almost 20 years earlier Thomson (1920) had compared his theory with Spearman's and stated that his own was, "in closer accord with theories in use in biology and in the study of heredity" (p. 190). This was reiterated in Brown and Thomson (1925, p. 192), which we quoted earlier.

How does biology judge? When Spearman, Thomson, and Burt met, spoke, and debated at a British Psychological Society symposium in April 1939, they agreed that "the burden of further proof is in large measure transferred to physiology" (Burt, 1939/1940, p. 86). The inescapable vagueness of Spearman's and Thomson's biological constructs and propositions, together with the limited advances that have been made since then in identifying the brain's units and which of the brain's parameters correlate with intelligence differences, mean that it is not yet, and perhaps never will be, possible to definitively judge which was the more correct. More than 30 years ago, an attempt was made to test Thomson's theory using EEG amplitude in groups of children who were good and poor readers (Maxwell, Fenwick, Fenton, & Dollimore, 1974). The parameters used in this study have not been replicated as reliably different in high- and low-ability individuals. However, the study was innovative in suggesting that high-ability people have generally more efficient brains. Also, it founded its hypothesis testing on the idea that greater Thomsonian sampling by a test might be indicated by its *g*-loading: Tests with greater *g*-loadings involve greater sampling (a point later made by Eysenck, 1987). This study and its ideas were discussed at length and developed by Haier (1993), who employed whole-brain glucose metabolic rate (as assessed using PET during mental work) as a measure of brain efficiency, and had found some evidence in small studies that people of greater intelligence did have more efficient brains (Haier et al., 1988).

Today, PET is used rarely in brain imaging studies of intelligence differences. There are other sources of brain variance that might be used to test between Thomson's and Spearman's ideas. For example, another biological parameter that might help is grey matter volume. It was suggested above that tests which have higher *g*-loadings might, according to Thomson's theory, involve sampling from a great range and number of bonds (Maxwell et al., 1974). Supportive of this is the finding that performance on tests with greater *g*-loadings correlated significantly with grey matter volume in more regions of the brain (Colom et al., 2006). Other brain parameters could be examined. If the vascular–neural–metabolic processes indicated by the BOLD response in magnetic resonance imaging (MRI) are generally more efficient in people with higher intelligence, then that would provide an account of some brains being generally better than others. The same would be true if there were general white matter differences throughout some brains, such that some were of better white matter integrity than others (Chiang et al., 2009; Yu et al., 2008). Chiang et al.'s (2009) study in twins found that brain white matter integrity is

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significantly correlated with intelligence in several brain regions, is heritable, and has a genetic correlation with intelligence. A general account might also be true at the synaptic level if, in some people, cognition-relevant multi-protein synaptic assemblies were more efficient in some than in others (Laumonnier, Cuthbert, & Grant, 2007).

On top of such a biological *g* it would be possible to accommodate group and more specific factors. Areas of specialization within the brain are well documented, for example, for spatial and language functions, and Anderson's theory of intelligence (Anderson, 1992) outlines such a plan. Variance below the level of *g* could also be sought in the specifics of some distributed networks, or at the level of specific types of neurotransmitter and receptor complexes (Grant, 2007).

Modern biology suggests other viable foundations for Thomson's model. For example, it is not yet known which constructs underlie the association between larger brains and higher intelligence. This could be individual differences in numbers of neurons, or dendritic processes and spines, any of which could provide Thomson's "elements" that allow more or fewer bonds to be formed and then sampled. In modern parlance, perhaps Thomson's *g* is the richness of the brain's distributed connectionist networks. Again, group and other factors could be accommodated, because it is known that the brain is not without structure, and there are defined neural networks whose variation can add variance on top of the basic overall richness of the elements. (Thomson also used the term "richness," but in a different sense, referring to tests; Thomson, 1939, p. 45). And the "elements" are not all of one type. Variation in the operation of say, dopaminergic or glutamergic or other specific functions could add variance that was additional to that of the overall network richness (Grant, 2007). Progressing to more ultimate causes, the recent realization that the genetic contributions to intelligence differences are probably many, uncorrelated and small, and likely to involve genes that influence systems (such as the vascular system) in addition specifically to those that affect the brain (Davis et al., 2008) accords with some of Thomson's (1924, p. 233) early guesses about the genetic "elements" of intelligence.

It is in Mendelism, where the idea of units of inheritance has proved so fruitful, that the idea of mental ability being dependent upon a number of unitary factors finds hope and support. The unitary factors, however, need not, as the above passage shows, be exclusively factors involving the quality and structure of the nervous system. Nor are they likely to coincide with any more or less unitary qualities commonly used in classifying the minds with which we are daily brought into contact, such as quickness or instability.

So Thomson might have got the genetic model of intelligence differences correct, if these were his intended units. But one problem with Thomson's biological guesses is that he places each-way bets, by sometimes thinking that units could be neurons, or connections, or neurone arcs, or genes. Burt (1940, p. 52, note 2) thought that there were problems here; for example, he thought that, as elements, genes might well be uncorrelated, but neurone arcs not. It is difficult to judge whether, if Thomson is correct in some details, this is not a Type I statistical error on our part.

The debate between Spearman and Thomson lasted almost 30 years. During all of that time Burt was active, theoretically and empirically, in the same field, but made less of an impact. In part, this is because Burt corresponded in a business-like way with both of them and tried to be diplomatic, emphasizing areas of agreement (e.g., Burt, 1939b, 1939/1940). This attempt to integrate rather than dispute is seen clearly in Burt's (1940) *The Factors of the Mind*. There are many examples there of his bringing together Spearman's and Thomson's ideas (e.g., p. 160), times when he disagrees with both (e.g., p. 215), and instances of sophisticated biological thinking. Thus, for example, "like Thomson, I consider that 'persons differ widely in the richness of their brains', but I lay more stress on the *systematization* of bonds (as he and Thorndike term them) than upon the mere *number* of 'bonds'" (p. 217; emphases in the original). This is not the place to reexamine Burt's ideas in detail, but such dynamic neural remodeling sounds quite up-to-date with respect to the development of intelligence differences and the brain changes that correlate with them (e.g., Shaw et al., 2006).

Burt saw, as we did above, that the Spearman–Thomson difference might be more apparent than real. Thus, "Spearman's basic components are rather like the organs of the body; Thomson's are more like its cells" (p. 11). And, if we focus for a moment on what Spearman and Thomson agreed upon—that mental tests correlate positively—then Burt (1940) saw that they might be hypothesizing about the same phenomenon but at different levels of analysis:

> (To put it crudely) a homogenous brain, consisting merely of a very large number of similar nerve cells, identical in nature and in strength, would obviously be a brain governed by a single general factor. In short, there is no mathematical difference between assuming only a single factor, varying continuously, and assuming an infinite (or indefinitely large) number of unit factors forming a single homogeneous "pool." A bushel of wheat is still a bushel, whether we call it corn or insist that it is composed of innumerable grains. (p. 160)

This statement echoes the mathematical analysis of Maxwell Garnett, though it is not clear whether Burt appreciated this point. When Garnett (1919/1920) had demonstrated the equivalence of Thomson's and Spearman's models he did so by showing that his general factor was, essentially, proportional to the sum of the contributions from all the bonds, a point acknowledged in Brown and Thomson (1925, p. 196). The general factor was thus the combined effect of all the bonds or, as Spearman saw it, his general factor was parceled out in small bits to the bonds.

11. Implications for the Meaning of Intelligence, and Beyond

The obvious conclusion of this work, which bears repetition, is that one cannot assume, as Spearman and many since appear to have done, that a good fit of a model is sufficient to explain the nature of what happens in the brain when test items are attempted. Neither does the existence of two or more equally good models mean that one of them must provide an adequate physical account. There may well be other models, not yet discovered, which account for the data equally well. The recent work of van der Maas et al. (2006) is an interesting step in that direction but, for reasons set out in Section 5, we do not think that it is yet in a state to account satisfactorily for the pattern of positive correlations that one finds when analyzing test results.

We have chosen to present the material in its historical context because it helps one to understand the thought processes by which

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the main protagonists came to their positions and to see how their attitudes differed. For his part, Spearman was never shifted from his attachment to the *g* hypothesis, whereas Thomson adopted a much more open and flexible approach to the extent that he was able to move easily between models on pragmatic grounds. This may go some way to explaining why Spearman's idea of continuous underlying factors has dominated factor analysis and structural equations modeling ever since. Our new work, showing that the bonds model can be extended to the multi-factor situation, shows that the two approaches can now compete on equal terms.

We quoted Thomson's view that biology should decide between his and Spearman's models, but our investigation of the contemporary position shows that the evidence still does not clearly come down in favor of either of them. In fact it strongly suggests that the true situation is more complicated than either model suggests. Our purpose here in putting the spotlight on Thomson's model is not because it is necessarily better than its main rival but because, by introducing the notion of "bonds," it may take us a little closer to the realities of the brain and so to a more adequate model of intelligence.

Beyond the historical interest, the focus on intelligence, and the interesting outcome that Thomson's and Spearman's models of intelligence are still both viable, there are at least two more avenues of research that can follow on from our account. First, the psychometric models of Spearman and Thomson—especially where they differ—may be used to formulate competing hypotheses in studies of the biology of intelligence. Support for Spearman's *g* could be sought in biological constructs to which a metric of quality may be applied, which differs between individuals, and which has a homogeneous value throughout a person's brain. This could point toward an examination of aspects of the brain's general fabric. A candidate in this field would be further work on brain white matter integrity, which, if further research finds it to be generally better in some people than others and also confirms the genetic correlation with intelligence (Chiang et al., 2009), would support Spearman's ideas. This is now possible to test empirically. Of course, separate sources of variance would then have to be sought for Spearman's specific (*s*) factors. Support for Thomson's theory would come from finding that within-person homogeneous aspects of brain fiber architecture did not correlate with intelligence. Positive support for Thomson's ideas could be sought by identifying brain units, and how they combine in the solving of mental problems. The prospect of this is probably not imminent. There is the extra problem of identifying sources of individual differences in performance, whether that is in the number of units, how they are sampled, their individual efficiency, and so forth.

Second, there should be recognition of the fact that intelligence is not the only field in psychology in which there are statistical models of mental performance and speculations about how these are founded in brain biology. Although it is beyond the scope of this article, we are grateful to an anonymous reviewer for pointing out the following idea: "[that] bonds are probabilistically recruited for solving specific tasks reminds me of current models of working memory that assume synchronized firing of model neurons binds together features that belong to the same item" (with references to Nairne, 1990; Oberauer & Kliegl, 2006; Raffone & Wolters, 2001). Individual differences in working memory are strongly correlated with intelligence, and memory differences more broadly form a domain of cognitive function with a high loading on *g* (Carroll, 1993). Such models of memory, therefore, are relevant to models of intelligence more generally. Therefore, within such models, also, we should seek discrete elements that are sampled (Thomson) and continua (Spearman). The models of working memory referred to above tend to emphasize units—such as neural assemblies and cells. And the model of Nairne (1990) involves sampling of elements with certain probabilities. However, if one were looking for a continuum, Raffone and Wolters's (2001) account includes a continuum time model for the potassium current in neurons.

In summary, we propose that Thomson's model of intelligence should be tested alongside Spearman's as still viable; that the parameters of brain structure and function that already are and are becoming available via neuroimaging should be especially used to test these hypotheses; and that broader integration should take place with other models of mental performance, beyond those addressing intelligence.

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Appendix

Derivation of the Covariance Matrix

Here we derive the covariance matrix for the generalized version of the bonds model given in Section 9.

When calculating expectations involving both **A** and **e** we first condition on **A** and thus obtain the expectation in two stages. Thus $E(\mathbf{A}_r\mathbf{e}_r) = EE(\mathbf{A}_r\mathbf{e}_r|\mathbf{A}_r) = 0$. Hence

$$E\mathbf{x} = E\sum_{r=1}^{k} E(\mathbf{A}_r\mathbf{e}_r|\mathbf{A}_r) = 0. \quad (1)$$

The covariance matrix is $E(\mathbf{x}\mathbf{x}')$, which is given by

$$E(\mathbf{x}\mathbf{x}') = E(\mathbf{A}_1\mathbf{e}_1 + \mathbf{A}_2\mathbf{e}_2 + \ldots + \mathbf{A}_k\mathbf{e}_k)(\mathbf{e}_1'\mathbf{A}_1' + \mathbf{e}_2'\mathbf{A}_2' + \ldots + \mathbf{e}_k'\mathbf{A}_k'). \quad (2)$$

The typical term in the expansion of this expression is

$$E\mathbf{A}_r\mathbf{e}_r\mathbf{e}_s'\mathbf{A}_s' = E[\mathbf{A}_rE(\mathbf{e}_r\mathbf{e}_s'|\mathbf{A}_r, \mathbf{A}_s)\mathbf{A}_s'], \ (r, s = 1, 2 \ldots k). \quad (3)$$

If, as we have assumed, the **e**s are mutually independent, the only terms in Equation 3 which are nonzero are those where $r = s$, in which case we have

$$EE(\mathbf{A}_r\mathbf{e}_r\mathbf{e}_r'\mathbf{A}_r') = E[\mathbf{A}_rE(\mathbf{e}_r\mathbf{e}_r'|\mathbf{A}_r)\mathbf{A}_r'] = E(\mathbf{A}_r\mathbf{V}_r\mathbf{A}_r'), \quad (4)$$

say, where $\mathbf{V}_r$ is the covariance matrix of $\mathbf{e}_r$. We also assume that the elements of $\mathbf{e}_r$ are mutually independent so that the matrix $\mathbf{V}_r$ is diagonal with typical diagonal element v_{rhh}. In that case, the (i, j)th element of $E(\mathbf{x}\mathbf{x}')$ is

$$\sum_{r=1}^{k}\sum_{h=1}^{N} E(a_{rih}a_{rjh})v_{rhh}.$$

The final step is to determine the expectations, $E(a_{rih}a_{rjh})$, which depend on the sampling rule. Because the *a*s are indicator variables, it follows that $a_{rih}a_{rih} = a_{rih}$. If $i \neq j$, a_{rih} and a_{rjh} are independent (meaning that the selection for item i is independent of that for item j), it follows that $E(a_{rih}a_{rjh}) = E(a_{rih})E(a_{rjh})$. But we assumed at the outset that the bond h was sampled, for the ith test at the rth pass, with probability p_{ri} (remembering that the probability is the same for all bonds). It therefore follows that

$$E(a_{rih}) = p_{ri}.$$

Using these results we deduce that

$$cov(x_i, x_j) = E(x_i\, x_j) = \sum_{r=1}^{k}\left\{\sum_{h=1}^{N} p_{ri}p_{rj}v_{rhh}\right\},$$

$$= \sum_{r=1}^{k} p_{ri}p_{rj}\,\text{trace}\,\mathbf{V}_r \ (i, j = 1, 2, \ldots, n, i \neq j). \quad (5)$$

When $i = j$, $E(a_{rih}a_{rih}) = E(a_{rih}) = p_{ri}$, and so

$$var(x_i) = \sum_{r=1}^{k} p_{ri}\,\text{trace}\,\mathbf{V}_r \ (i = 1, 2, \ldots, n). \quad (6)$$

Finally, therefore, if we further require that all the matrices, $\mathbf{V}_r$, have the same value, $\mathbf{V}$, then

$$\text{corr}(x_i, x_j) = \frac{\sum_{r=1}^{k} p_{ri}p_{rj}}{\sqrt{\sum_{r=1}^{k} p_{ri}\sum_{r=1}^{k} p_{rj}}} = \sum_{r=1}^{k}\left(\frac{p_{ri}}{\sqrt{\sum_{r=1}^{k} p_{ri}}}\,\frac{p_{rj}}{\sqrt{\sum_{r=1}^{k} p_{rj}}}\right)$$

$$= \sum_{r=1}^{k} \lambda_{ri}\lambda_{rj}, \ (i, j = 1, 2, \ldots, n) \quad (7)$$

say. This has exactly the same form as the covariance for the linear factor model originating with Spearman as we intended to show.

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